

Ronit Amar Bhatia

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Education

Cornell University Master of Engineering in Engineering Management	Aug 2024 – May 2025
University of California, Davis Bachelor of Science in Computer Science Minor in Technology Management	Sept 2020 – June 2024

Experience

Forward Deployed Engineer , Y Meadows	Jan 2026 – Present
- Automated document processing pipelines for 2500+ weekly documents, saving 100+ work hours a week for clients. - Deployed weekly automation workflows for 2–3 enterprise clients, integrating Gmail, Slack, and client-specific systems to eliminate manual operational tasks. - Built an internal RAG-based workflow assistant indexing 50+ runbooks, reducing team query resolution time by ~40%.	
Machine Learning Engineer Intern , QAlienAI (Remote)	Oct 2025 – Feb 2026
- Developed an LLM-driven compliance system using semantic similarity and rule-based checks for regulatory review. - Designed multimodal inference pipelines using AWS Bedrock and AssemblyAI, increasing OCR and ASR accuracy by 15%. - Engineered a scalable semantic search layer with pgvector, optimizing embedding generation and hybrid retrieval to enable high-precision access to regulatory text.	
Software Engineer Intern , Gallox Semiconductors (Remote)	Nov 2024 – Jan 2025
- Developed automation solutions for power device characterization, improving testing efficiency by 35%. - Engineered a real-time data visualization system in Python, accelerating debugging and analysis during lab evaluations. - Developed Python APIs to interface with lab instruments, automating data acquisition and control during device testing.	
Student Research Assistant , Cornell CALS	Aug 2024 – Dec 2024
- Utilized k-means clustering and regression to analyze crop-level GHG emissions, improving pattern detection by 30%. - Built Python data pipelines to preprocess geospatial crop emissions data, enhancing feature consistency for ML models. - Implemented clustering models to group crops by emission intensity, enabling targeted mitigation strategies.	

Projects

Treadwell: On-Device Hazard Navigation [GitHub] [Demo]	Aug 2026
- Developed a multimodal hazard-detection pipeline using Swift, Gemma 4, and LiteRT-LM with zero cloud inference. - Engineered distributed edge inference between an iPhone camera and a ~2.4 GB model on an M3 Pro using Bonjour/TCP. - Reduced latency from 5–7s to 1–2s; implemented JSON validation, confidence gating, deduplication, and 20 XCTest cases.	
UnClogAI: AI-Powered Workflow Bottleneck Detector [GitHub]	Aug 2025 – Sept 2025
- Developed an AI workflow automation tool using graph analytics to detect bottlenecks and improve process efficiency. - Designed a multi-agent system with LangGraph and Hugging Face LLMs for predictive insights and risk assessment. - Optimized data pipelines and graph computations to enable efficient, real-time workflow analysis at scale.	
Excellensight: AI-Powered Feedback Analyzer , Cornell University [GitHub]	Mar 2025 – May 2025
- Designed a CNN-BiLSTM-Attention architecture for review classification and insight generation without pre-trained LLMs. - Built a full NLP pipeline covering preprocessing, training, evaluation, and automated Markdown/HTML reporting. - Achieved over 98% validation accuracy on 10K+ reviews; reports include key trend summaries and visualizations.	
Taskify: AI-Powered Task-to-Team Member Matching [GitHub]	Mar 2025 – May 2025
- Built a Transformer model fusing embeddings and structured data to predict task-member fit with 89.6% accuracy. - Generated and labeled a synthetic dataset (40K+ samples) using sentence-transformers and feature engineering. - Deployed a real-time Streamlit app to recommend top-ranked teammates with confidence scores and reasoning logic.	

Technical Skills

Programming Languages: Python, C/C++, SQL, Go, JavaScript, HTML, CSS, Lisp, Prolog, MicroPython

Machine Learning & AI Tools: TensorFlow, PyTorch, Scikit-learn, LangChain, Ollama, LangFuse, LangGraph, Strands Agents, AWS Bedrock, AssemblyAI, Resend

Tools & DevOps: Next.js, React, Node.js, Supabase, Clerk, TanStack Query, Docker, GitHub, GitHub Actions, VSCode, MATLAB, JIRA, Power BI

Cloud Platforms: Vercel, AWS (S3, EC2, Lambda, SageMaker, RDS, DynamoDB), GCP (BigQuery, Vertex AI, Cloud Run)